AI & Industry-Focused Statistics Interview Questions

1. AI Chatbot Evaluation

Your organization deploys an AI customer support chatbot. Management claims it has an **85% accuracy** based on internal testing.

After deployment, customer complaints increase.

Questions

- Is accuracy alone sufficient to evaluate the chatbot?
- What additional statistical metrics would you analyse?
- How would you determine whether the drop is statistically significant or simply due to random variation?

Concepts Tested

- Sampling
- Confidence intervals
- Statistical significance
- Evaluation metrics

2. Prompt A/B Testing

Your AI team has created two prompts for summarising customer complaints using an LLM.

Prompt A receives an average human rating of **4.3/5**.

Prompt B receives **4.5/5**.

Management immediately wants to switch to Prompt B.

Questions

- Would you recommend deployment immediately?
- What statistical tests would you perform?
- What sample size considerations are important?

Concepts

- A/B Testing
- Hypothesis Testing
- Confidence Interval
- Statistical Power

3. AI Hallucination Monitoring

A GenAI application generates product descriptions.

Last month:

Hallucination Rate = 6%

This month:

Hallucination Rate = 8%

The AI Product Manager asks whether the model quality has deteriorated.

Questions

- Can you conclude quality has decreased?
- What statistical evidence would you require?
- How would confidence intervals help?

4. LLM Response Quality

Three different LLMs generate answers for customer queries.

Model A

Average Score = 4.6

Model B

Average Score = 4.5

Model C

Average Score = 4.7

Human evaluators score each response.

Questions

- Which statistical method would you use to determine whether the difference is meaningful?
- Why wouldn't comparing averages alone be sufficient?

Concepts

- ANOVA

- Hypothesis Testing
- Variance

5. Feature Importance vs Correlation

An ML model predicts customer churn.

The feature **Number of Support Tickets** has the highest importance.

A stakeholder concludes:

"Support tickets cause churn."

Questions

- Is this conclusion statistically valid?
- How would you explain correlation, feature importance, and causation?

6. AI Recruitment Bias

A hiring model shortlists:

82% of male candidates

64% of female candidates

The HR team claims the model is biased.

Questions

- What statistical analysis would you perform before reaching that conclusion?
- What additional information would you require?
- How would sample size affect your decision?

7. AI Recommendation Engine

A recommendation model increases click-through rate by 2%.

However,

Average purchase value decreases by 5%.

Questions

- Which metrics should drive the business decision?
- How would statistics help determine whether this trade-off is acceptable?

8. Drift Detection

Your fraud detection model achieved 96% accuracy during training.

Six months later accuracy drops to 84%.

Questions

- How would statistics help determine whether this is:
    - Data Drift
    - Concept Drift
    - Sampling Bias
- What statistical distributions would you compare?

9. LLM Benchmarking

Your company wants to compare GPT, Claude, Gemini and an open-source model for document summarisation.

Each model summarises 1,000 documents.

Questions

- How would you design the experiment?
- Which statistical tests would you use?
- How would you eliminate evaluator bias?

10. AI Agent Performance

An AI agent automates invoice processing.

Last month

Success Rate = 91%

This month

Success Rate = 94%

Management says the AI has improved.

Questions

- Is a 3% improvement enough to claim success?
- How would you statistically validate this?

Modern Data Analyst Scenarios

11. Dashboard Anomaly Detection

A Power BI dashboard shows revenue increasing by 25%.

Later you discover duplicate invoices were loaded.

Questions

- Which descriptive statistics would have helped identify this?
- What validation checks should every analyst perform before publishing dashboards?

12. Customer Segmentation

A retail company wants to segment customers for personalised marketing.

Questions

- Why is standard deviation important before clustering?
- Why is feature scaling necessary?
- What problems occur if one variable dominates the others?

13. AI Sales Forecasting

An ML model predicts next month's sales.

Actual sales differ by only 1%.

The business celebrates.

Questions

- Does one successful prediction prove the model is reliable?
- What statistical evaluation would you perform over several months?

14. Healthcare AI

An AI model detects diabetic retinopathy.

Accuracy = 96%

Recall = 71%

Doctors are concerned.

Questions

- Why is recall critical here?
- Which error is more dangerous?
- Would you deploy this model?

15. LLM Fine-Tuning

After fine-tuning an LLM,

BLEU Score increases by 2%.

ROUGE Score decreases by 1%.

Human ratings remain unchanged.

Questions

- Has the model actually improved?
- Which evaluation should carry the most weight?
- Why?

16. AI Fraud Detection

Fraud rate is only 0.15%.

Your model achieves 99.8% accuracy.

Questions

- Why might accuracy be misleading?
- Which metrics would you present to leadership?
- How would class imbalance affect model evaluation?

17. Recommendation System Experiment

Netflix-style recommendations increase viewing time by 8%.

Customer retention remains unchanged.

Questions

- Which KPI should determine success?
- How would you statistically validate the business impact?

18. Generative AI Cost Optimisation

A company switches from GPT-4 to a smaller open-source model.

Cost reduces by 70%.

User satisfaction drops from 4.8 to 4.6.

Questions

- Would you recommend the change?
- What statistical evidence would support your recommendation?

19. AI Data Quality

An LLM is trained on historical customer support tickets.

Later you discover:

- 18% duplicate records
- 12% missing labels
- inconsistent categories

Questions

- How might these issues affect model performance?
- Which statistical techniques would help identify these problems before training?

20. Executive Decision-Making

The CEO asks:

> "Can you guarantee that deploying this AI model will increase revenue?"

Questions

- How would you respond using statistical reasoning?
- Why do data scientists talk about confidence rather than certainty?
- How would you communicate uncertainty to business stakeholders?